GEOMATICS
NTNU

TBA4245 GEODESY

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Example exercise no. 1: GPS baselines. Baseline transformation and accuracy measures
Solution

The page references are to the compendium «Basic GPS» («Grunnleggende GPS») on Blackboard.

Task no. 1. Transformations of satellite baselines

Here the method and calculations are shown for the baseline Solemsvåttan-Holtermannsveien (S-H). Similar calculations for other baselines.

(a) Calculate the local geodetic coordinates of the GPS baselines from Solemsvåttan to Holtermannsveien and from Ladehammeren to Holtermannsveien. (Conversion to a local terrestrial horizontal coordinate system.)
Which easy/simple calculations can be done to check if the conversion is OK? Carry out this check.

The GPS baselines are originally in the GPS datum WGS84. We want to convert the GPS baseline Solemsvåttan-Holtermannsveien from a Cartesian XYZ system to a local terrestrial horizontal coordinate system with origin in Solemsvåttan.

Problems:
Which values of the latitude and the longitude shall be used for the starting point of the baseline? The datum of the observations? The datum of the final coordinates and heights?

We need the latitude (φ) and longitude (λ) of point Solemsvåttan, preferably in the datum WGS84. **Solution:** We can use the φ and λ calculated from the given EUREF89-UTM coordinates. The shape of the ellipsoids (half axis a and b) and the way the ellipsoids are linked (oriented) to the Earth is the same for EUREF89 and WGS84. Therefore latitude and longitude in both these datums will work, as we only need approximate values. Eventually systematic differences of bearings and scale between the baselines and the known points can be eliminated or reduced by using rotation and scale as unknowns in the adjustment/calculation of the UTM coordinates.

One of the methods:

Use the official transformation program from the Norwegian mapping authority, SkTrans. We can go directly from the UTM projection of Euref89 (*N(orth)* and *E(ast)*) to Euref89 values (φ and λ) and Cartesian X, Y and Z with SkTrans. Transfering (or more correct converting) the values of the start of the baseline, **point S**(olemsvåttan):

Point S: From system: EUREF89, UTM axis 32: N:7032158.662 E:579058.469
To system: EUREF89-Geo Latitude: 063° 24’ 33.1032” = 63,40919533° (degrees)
Longitude: 010° 34’ 59.2253 “= 10,58311814°

The transformation equation (10.15c) on page 58 in "Basic GPS" gives the coordinates:

The transformation matric:

$$\mathbf{T} = \begin{pmatrix} -0,879014859 & -0,164235107 & 0,44761558 \\ -0,183661727 & 0,982989507 & 0 \\ 0,440001418 & 0,08220985 & 0,894226086 \end{pmatrix}$$

Local coordinates of point H with origin in S:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}_{LG} = \begin{pmatrix} 520,6809 \\ -9255,4658 \\ -358,3410 \end{pmatrix}$$

Simple checks:

1: An easy/simple calculation to check if the conversion is OK:

Calculate the length of the baseline (= slope distance) both from the local coordinates and the 3 baseline components (the Cartesian ΔX, ΔY, ΔZ). Should be exactly the same:

S_{slope} = 9277,0234 m

NB: This check is not 100 % thorough. For instance will errors on the latitude and/or longitude give an OK result (same lengths) but nevertheless erroneous coordinates.

2: Closure, sum of ΔX in the triangle L-H-S-L should be close to 0, same for ΔY and ΔZ.

From	To	ΔX	ΔY	ΔZ
L	H	3541,713	-811,048	-1666,943
S	H	1084,518	-9212,999	-87,373
S	L	-2457,183	-8401,958	1579,590
	Sum:	-0,012	+0,007	-0,020

Example: Sum ΔX in triangle S-L-H-S = -2457,183 + 3541,713 – 1084,518 = –0,012 m

The 3D gap is obtained by: 3Dgap = sqrt($0,010^2 + 0,007^2 + 0,020^2$) = 0,0244 m = 24,4 mm

Comparing the gap with the apriori (assumed) accuracy of a baseline:

5 mm + 0,5 mm/km in the plane and 7,5 mm + 0,75 mm/km in height

Accuracy of each of the baselines, in millimeter:

From	To	Distance	Sd, in plane	Sd, in height	3D accuracy
S	H	9277,023	9,64	14,46	17,38
S	L	8895,266	9,45	14,17	17,03
L	H	3997,528	7,00	10,50	12,62

Accuracy of the 3 baselines forming a triangle: = sqrt($17,38^2 + 17,03^2 + 12,62^2$) = 27,4 mm, which is larger than the 3Dgap of 24,4 mm

(b)	Calculate the distance and the bearing from Solemsvåttan to Holtermannsveien in the map projection plane. Compare with values calculated directly from the EUREF89 UTM coordinates.

Calculating the distance and the azimuth form S to H from the local coordinates (LG) in (a):

(10.18)-equations in “Basic GPS”:

α = atan (y / x) =-96,4223657876 gon+400 = 303,577635 gon

$S_{\text{slope}} = \sqrt{}(x^2 + y^2 + z^2)$ = 9277,0234 m

$S_{\text{Horizontal}} = \sqrt{}(x^2 + y^2)$ = 9270,1001 m

As we are on the ellipsoid, the radius of the curvature is dependent of the latitude, the chosen ellipsoid and the azimuth of the baseline. We need a good value of the radius of the curvature. The formulas in the compendium “Geodesy part 1” gives the radius of the curvature in point S in the direction S-H: $R_{\text{S-H}}$ = 6395249,99 m (calculated from the two principle radii of curvature R_{M} = 6386653,19 m (along the meridian) and R_{N} = 6395277,24 m (east-west direction)).

Using equations in chapter 11 in compendium «Basic GPS» or in chapter 2 in “Geodesy part 1”.

(1) Distance in the UTM map projection plane.

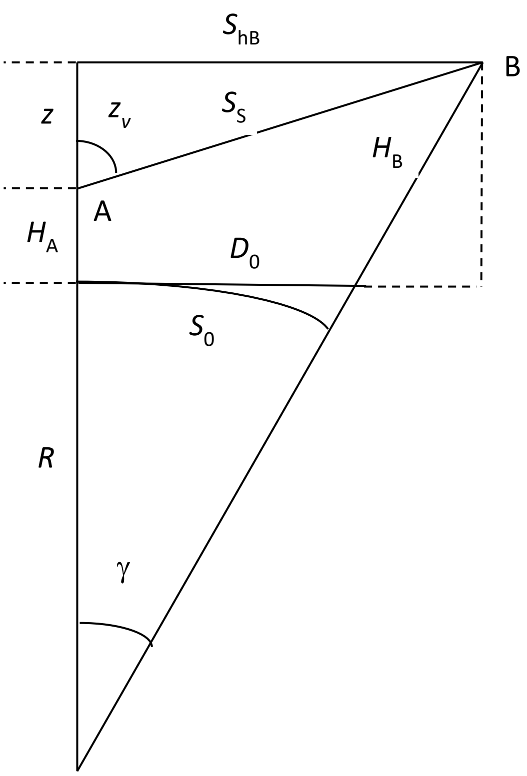
See figure 11.1 on page 59 (the left figure below). The figure to the right is the actual figure (points S and H).

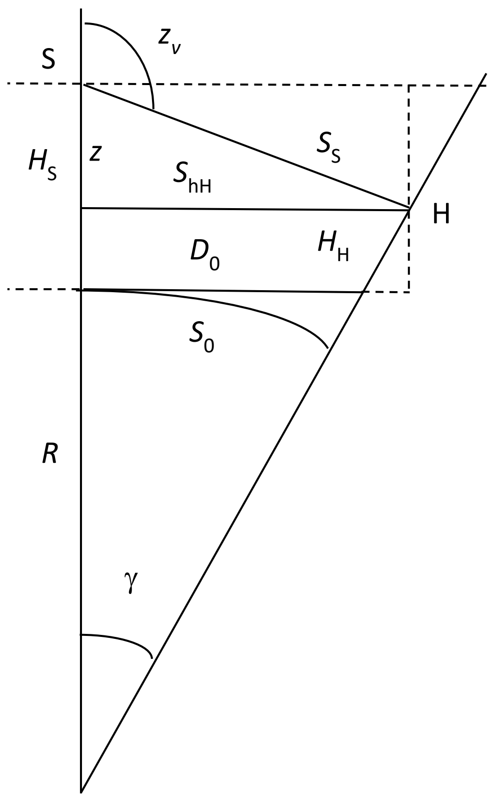
First we calculate the angle γ in the center of the sphere with radius R, the radius of the curvature in S in direction of H. S_{hH} is the horizontal distance in the height H_{H} for Holtermannsveien, is 9270,1001 (from the 10.18 equations, see above). H_{S} is the ellipsoidal height of S (we go directly to the ellipsoid, not calculating the distance on the geoid). z is a coordinate of H in the LG system, is negative here, as the origin is in point S:

$$\gamma = a\tan\left(\frac{S_{hH}}{R + H_S + z}\right) = a\tan\left(\frac{9270{,}1001}{6395249{,}99 + 462{,}071 + (-358{,}341)}\right) = 0{,}001449504437 \text{ rad}$$

Distance on the ellipsoid:

By using the equation (11.3) with ellipsoidal height we obtain the distance in the height H = 0, on the ellipsoid: $S_{0\_\text{ELL}} = R \cdot \Upsilon$ = 9269,9432 m (11.3), is ellipsoidal distance.





Correction from the ellipsoid to the map projection plane:

General equation is (4.60) in “Geodesy part 2”. See also about the UTM equation that has to be used here, as UTM map projection is used.
UTM equation where the ellipsoidal term in (4.60) is left out, using a sphere equation as the distance is relatively short (ell. term = 0).

$$\boxed{S_M = S_0 + \frac{S_0}{6R^2}\left(y_A^2 + y_B^2 + y_A \cdot y_B\right) - 0{,}0004 \cdot S_0}$$

The UTM traps:

1. The last term is because of the scale factor of 0,9996 for the UTM map projection. Gives a negative correction on the distance of 40 cm per km, or: -0,0004 · S
2. The east value y is the distance from the north axis. The factor 0,9996 and the offset of 500000 m for the UTM E(ast) value gives: $y = (E - 500000)/0{,}9996$

Correction from S_0 til the distance in the UTM map projection plane, D_{M1}:

ΔS == +0,6291 – 3,7080 = -3,0789 m.

Distance in the UTM map proj. plane: $D_{M1} = S_{0\_ELL} + \Delta S$ = 9269,9432 – 3,0789 = 9266,8643 m

Distance calculated from the EUREF89 UTM coordinates: D_{M2} = 9266,8696 m
A difference of 0,0053 m.

Comment: A real situation, as the GNSS baseline was measured and the points and their values are found in the GNSS network of the Trondheim municipality. Shows that the quality is very good.

The old version of this example exercise used coordinates of the points measured with classical instruments and values in the old datum NGO1948. The difference was 0,085 m, says something about the improvements of the coordinates when using GNSS (GPS) ….

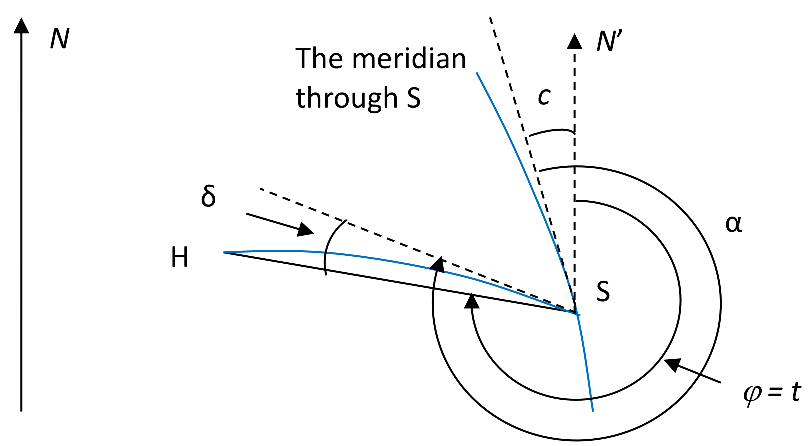
(2) Bearing (*retningsvinkel*) in the UTM map projection plane

Starting with the azimuth calculated from the equation (10.18), see a previous page:

Azimuth from S to H: α = 303,577635 gon

The azimuth is the angle from the meridian through S and to the line S-H on the ellipsoid.
See the figure below for how these two lines are mapped in the plane, as curved (blue) lines.
The bearing, $\varphi = t$, is the angle between a parallel to the north axis and the straight line S-H.

⇨ Corrections must be applied to both the curved lines, see below:



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- **The meridian convergence in S, *c***. Several ways, remember to use radians in the equations and use the difference between the longitude of point S and the longitude of the north axis in the UTM zone 32 (= 9 degrees) as l in the equations, l = 10,58311814° – 9° = 1,58311814°. The values of latitude and longitude were calculated in question (a):

  $c = l \times \sin B + (l^3/3) \times \sin B \times \cos^2 B \times (1 + 3 \times \eta^2 + 2 \times \eta^4) + (l^5/15) \times \sin B \times \cos^4 B \times (2 - \tan^2 B)$

  or

  tan c = equation in chapter 4 in «Geodesy part 2»

  Using the c = equation above. Denomination: In the equation above it might be easiest to use radians for the latitude l, as we have l^3 and l^5 in the terms. Calculate c in radians and then transform the radian value to gon (as we are using gon in the map projection plane).

  $$c_S = 1{,}5729617 + 0{,}0000805 - 0{,}00000001 = 1{,}5730422 \text{ gon}$$

- **The map projection correction, δ.** From the ellipsoid to the map, δ, equation (4.59) in Holsen "Geodesy part 2", or equation (11.8) in "Basic GPS". The sphere equation is Ok to use here, as the distance is relatively short.

  δ = -0,000 0173 gon. (Low value as the line is going ca. east-west)

Bearing in the UTM plane from the GPS baseline, we use the figure to see the sign of the correction terms:

$$\varphi = t = \alpha - |\delta| - |c| = 303{,}5776346 - 0{,}0000173 - 1{,}5730422 = 302{,}004575 \text{ gon}$$

Bearing calculated from the given UTM coordinates: φ = 302,004568 gon

A difference of 0,007 mgon

An angle of 0,007 mgon gives an arc length (tverravvik) of 1 mm on the distance S-H: …Checking a GNSS network with GNSS…, see the comments on the distance above.

(c)	**Either: A:** Calculate the zenith angle from Solemsvåttan to Holtermannsveien from given heights and coordinates. Compare with the zenith angle calculated from the GPS measurements. **Or, probably easier: B:** Calculate ΔH (orthometric height difference) from the GPS measurements and compare to the ΔH in the table above.

Method: Calculating ΔH in the datum NN2000, and then comparing with the given NN2000 heights of S and H.

Starting point: Using equations 10.18 in "Basic GPS" and calculate the ellipsoidal zenith angle z_{v_LG} and the ellipsoidal height difference ΔH_{LG} from the local horizontal coordinates in (a). The zenith angle and ΔH need not to be corrected for refraction.

Z_{v_LG} = 102,459666 gon $\qquad$ $\Delta H_{LG} = z$ = -358,3410 m

A: Correction from ellipsoid normal to the plumb line, the deflection of the vertical

We have to correct for the deflection of the vertical («loddavvik») as we start with ellipsoidal height (along the ell. normal) and end with a height measured along the plumb line («loddlinja») as this height is related to the geoid (mean sea level).

The final calculation of the coordinates and heights is done in the map projection plane. In the solution we therefore are using plane bearing (not azimuth) and distances in the map projection (not on the ellipsoid). Thus it is also easier to compare with the results from the land surveying program Gisline.

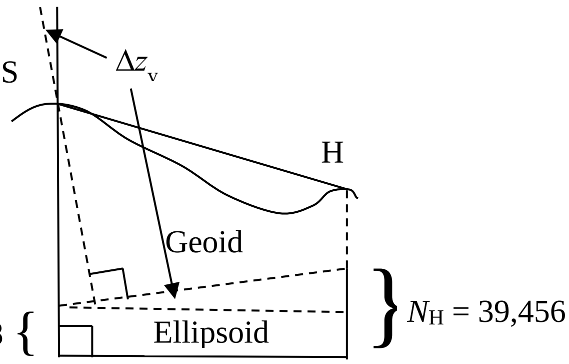
Two methods:
1. Calculate directly the correction for the actual line from the given values of the ellipsoidal heights and the NN2000 heights
2. Calculate by using the given values of the north and east components of the deflection of the vertical, ξ and η. See the frame below for how they are calculated (this is a topic in the course TBA4236 Theoretical geodesy)

Formulas: $\Delta z_v = \cos\alpha \cdot \xi + \sin\alpha \cdot \eta$ $\qquad z_{NN2000} = z_{ELL} - \Delta z_v = z_{ELL} - (\cos\alpha \cdot \xi + \sin\alpha \cdot \eta)$

Method no. 1: Calculating directly the component of the deflection in the actual direction, here S-H, by using the known ellipsoidal heights and the NN2000 heights.

Point	*North*	*East*	*Hell*	H_{NN2000}	*N*	Δx local	Δy local
S	7032158,662	579058,469	462,071	422,853	39,218	0	0
L	7036209,878	571150,913	108,875	69,634	39,241	4051,216	-7907,556
H	7032450,406	569796,193	110,462	71,006	39,456	291,744	-9262,276

The geoid-ellipsoid separations (geoid heights, geoidehøyder), *N*, are calculated in the table above. $N = H_{Ellipsoid} - H_{NN2000}$



Difference in geoidal height (geoid-ellipsoid separation):

$\Delta N_{S\text{-}H} = N_H - N_S = 39,456 - 39,218 = +0,238$ m

The correction Δz_v: $\qquad N_S = 39,218$

$\tan \Delta z_v = (N_H - N_S) / D_{S\text{-}H} = 0,238 / 9269,943$

=> $\Delta z_v = -1,634$ mgon (–, as *N* increases towards H). To milligon with $(200000^{mgon} / \pi)$

Method no. 2: From the given values of the components of the deflection of the vertical (loddavvikskomponentene):

In the actual area (points S, H and L) the components of the deflection of the vertical are given in the assignment text:

$\xi = +3,02$ mgon in the northern direction and $\eta = +1,73$ mgon in the eastern direction.

Using equation (12.5) page 70 in «Basic GPS»: $\Delta z_v = \cos\alpha \cdot \xi + \sin\alpha \cdot \eta$

The plane bearing from S to H from (a): $\varphi = 302,004575$ gon

The correction: $\Delta Z_{Defl} = \Delta Z_V = 0,095 - 1,729 = -1,634$ mgon

Calculation of the north and east components of the deflection of the vertical, ξ and η, by only using the values of the coordinates and heights given in the table above will give a geoid height model. As shown in the course TBA4236 Theoretical geodesy a first order geoid height polynom model is written as:

$N = -\xi \cdot \Delta x - \eta \cdot \Delta y + N_0$

where Δx and Δy are local coordinates with origin in point S. See the table for values.

Three equations of N (one for each point) and three unknowns (ξ and η and N_0):

In S: $N_S = 39{,}218 = -\xi \cdot 0 - \eta \cdot 0 + N_0$

In L: $N_L = 39{,}241 = -\xi \cdot 4051{,}216 - \eta \cdot (-7907{,}556) + N_0$

In H: $N_H = 39{,}456 = -\xi \cdot 291{,}744 - \eta \cdot (-9262{,}276) + N_0$

N_0 is calculated from the equation in point S: $N_0 = 39{,}218$ m

The two components ξ and η is calculated from the equations in points L and H:

$\xi = 3{,}02$ mgon Gisline value: 3,05 mgon

$\eta = 1{,}73$ mgon Gisline value: 1,70 mgon

Formula: $\Delta z_v = \cos\alpha \cdot \xi + \sin\alpha \cdot \eta = \cos\alpha \cdot 3{,}02 + \sin\alpha \cdot 1{,}73 = -1{,}634$ mgon

=> same result as method no. 1. (α is the plane bearing)

The difference of the results of the two methods:

The given values of ξ and η are calculated in the land surveying program Gisline as a part of an estimation of these two components and the coordinates and heights. This least square method of estimating the unknowns will give an optimal solution of all the unknowns involved.

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B: ΔH needs not to be corrected for refraction

C: Correction for the curvature of the earth

We have to correct for the curvature of the earth. Two methods:

(a) Calculating the correction in meters

(b) Correcting the zenith angle

Both equations are given in text books in basic land surveying (for instance «Grunnleggende landmåling»).

(a) Calculating the correction in meters.

$$\Delta H_{\text{Earth}} = \frac{\left(S_{slope} \cdot \sin z_v\right)^2}{2R} = \frac{\left(9277{,}023 \cdot \sin 102{,}459666\right)^2}{2 \cdot 6395249{,}99} = 6{,}7186 \text{ m}$$

(b) Correcting the zenith angle

Formula: $$\Delta Z_{\text{Earth}} = \frac{S_{slope} \cdot \sin z_v}{2R} \cdot \rho = 0{,}0461399 \text{ gon}$$

Calculate ΔH and H in the datum NN2000

Orthmetric height, or more correctly a normal height. Use a figure!

Two methods:

(a) By calculating the correction directly in meters
(b) By calculating the corrections by using the zenith angle

(a) Calculating the correction in meters

Δz_v = –1,634 mgon (from method no. 1)

Measured ellipsoidal ΔH from the equations (10.18) in Basic GPS: -358,3410
Correction, deflection of the plumb line: $\Delta H_V = D_{S\text{-}H} \cdot \tan \Delta z_v =$ -0,2380
Correction, the curvature of the earth: $\Delta H_E =$ +6,7186
Measured ΔH_{NN2000} of point H: ΔH = <u>-351,8604 m</u>

(b) Calculating the corrections by using the zenith angle

The sign of Δz_{Earth} is given for instance in equation on page 152 in “Grunnleggende landmåling” (2014), the sign of the term Δz_{DEFL} is given by the equation (12.5) page 70 in “Basic GPS”:

$$zv\_{NN2000} = z_{LG} - \Delta z_{Earth} - \Delta z_{Defl} = 102{,}459666 - 0{,}0461399 - (-0{,}001634)$$
$$zv\_{NN2000} = 102{,}4151597 \text{ gon}$$

NN2000 height difference (orthometric height difference) calculated from the GPS measurements, equation from text books in basic land surveying (Grunnleggende landmåling):

$$\Delta H = S_{H=H\_mean} / \tan Z_{ORT}) = (S_{Ell} \cdot (1 + H_{Mean} / R)) / \tan z_{v\_NN2000}) =$$
$$\Delta H = \{9269{,}9432 \cdot (1 + \{462{,}071+110{,}462) / 2\} / 6395249{,}99)\} / \tan 102{,}4151597) = \underline{-351{,}8606\text{m}}$$

Comparison measured and given height differences:

NN2000 height difference calculated from the GPS measurements: ΔH = -351,861 m
NN2000 height difference from the known/given heights: ΔH = 71,006 - 422,853 = -351,847 m
A difference of: 0,014 m

A difference of 14 mm over a distance of 9,26 km must be OK. Check with the empirical accuracy of a measured GNSS baseline.

The old version of this example exercise used heights of the points measured with classical instruments and values in the old datum NN1954. The difference was 0,045 m, says something about the improvements of the heights when using GNSS (GPS)….

Task no. 2. Accuracy measures of the measured GPS baselines

Calculations, using the baseline S-H.
Some of the calculations are the same as in task no. 1, for instance finding the transformation matrices.

(a) The 3D Cartesian *X*, *Y*, *Z* coordinate system

Calculate the values of the variance/covariance matrix of ΔX, ΔY, ΔZ of the baseline S-H.

For the values of the correlation matrix **K** and the standard deviations on ΔX, ΔY, ΔZ in the Cartesian geocentric coordinate system, see the assignment text.

Using mm, not meter, as denomination to obtain more practical values (closer to 1) in the calculations.

Calculating the variance/co-variance matrix, $\mathbf{C}_{CTRS}$, using equation (10.2) in the text book «Basic GPS». In the geocentric coordinate system, denomination is mm^2:

$$\mathbf{C}_{CTRS} = \mathbf{S}\cdot\mathbf{K}\cdot\mathbf{S} = \begin{pmatrix} s_{\Delta X} & 0 & 0 \\ 0 & s_{\Delta Y} & 0 \\ 0 & 0 & s_{\Delta Z} \end{pmatrix} \cdot \begin{pmatrix} 1,0000 & 0,1307 & 0,5335 \\ 0,1307 & 1,0000 & 0,1441 \\ 0,5335 & 0,1441 & 1,0000 \end{pmatrix} \cdot \begin{pmatrix} 0,5 & 0 & 0 \\ 0 & 0,4 & 0 \\ 0 & 0 & 1,0 \end{pmatrix}$$

$$\mathbf{C}_{CTRS} = \begin{pmatrix} s^2_{\Delta X} & \sigma_{\Delta X \Delta Y} & \sigma_{\Delta X \Delta Z} \\ \sigma_{\Delta X \Delta Y} & s^2_{\Delta Y} & \sigma_{\Delta Y \Delta Z} \\ \sigma_{\Delta X \Delta Z} & \sigma_{\Delta Y \Delta Z} & s^2_{\Delta Z} \end{pmatrix} = \begin{pmatrix} 0,25000 & 0,02614 & 0,26675 \\ 0,02614 & 0,16000 & 0,05764 \\ 0,26675 & 0,05764 & 1,00000 \end{pmatrix}$$

Notice that the co-variances are different from zero: Correlation exist between the three observations ΔX, ΔY, ΔZ. Why?

(b) From the 3D Cartesian X, Y, Z to the local geodetic coordinate system, x, y, z
Calculate the values of the variance/covariance matrix of x, y, z for the baseline S-H.

Transferring the variance/co-variance matrix, $\mathbf{C}_{CTRS}$, to the variance/co-variance matrix, $\mathbf{C}_{LG}$, in the local horizontal coordinate system with origin in the starting point of the baseline, S(olemsvåttan).

Using the error propagation law on the transformation equation (10.15) to obtain equation (11.12):

$$\mathbf{C}_{LG} = \mathbf{T}\cdot\mathbf{C}_{CTRS}\cdot\mathbf{T}^T$$

The transformation matric, found in equation (10.15 and 10.16):

$$\mathbf{T}_{\text{Solemsvåttan}} = \begin{pmatrix} -0,879014859 & -0,164235107 & 0,44761558 \\ -0,183661727 & 0,982989507 & 0 \\ 0,440001418 & 0,08220985 & 0,894226086 \end{pmatrix}$$

$\mathbf{C}_{LG}$ In the local horizontal coordinate system with origin in the starting point S, $(x, y, z)_{LG}$.
Notice the relatively low values of the elements of the variance/covariance matrix, denomination mm^2:

$$\mathbf{C}_{LG} = \mathbf{T}\cdot\mathbf{C}_{\mathbf{CTRS}}\cdot\mathbf{T}^{\mathbf{T}} = \begin{pmatrix} s^2_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & s^2_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & s^2_{zz} \end{pmatrix} = \begin{pmatrix} 0,18700 & -0,00384 & 0,13416 \\ -0,00384 & 0,15360 & 0,01049 \\ 0,13416 & 0,01049 & 1,06940 \end{pmatrix}$$

Calculating the standard deviations on x, y, z of the baseline S-H from the variances in the main diagonal: $s_x = 0{,}43$ mm $s_y = 0{,}39$ mm $s_z = 1{,}03$ mm   <- Unrealistic low values?

(c) From the local geodetic coordinate system, x, y, z to the decomposed observations in the UTM map projection plane and height difference or zenith angle referred to the geoid (the datum NN2000)

State with equations how the variance/covariance matrix of the decomposed observations of a GPS baseline (distance, bearing, height difference or zenith angle) can be calculated.

Calculate the values of the variance/covariance matrix of the decomposed observations for the baseline S-H.

Calculating the horizontal and slope distance, azimuth, the ellipsoidal zenith angle and the ellipsoidal height difference by using equation (10.18), is also done in question no. 1:

Slope distance, S_S	9277,0234 m
Azimuth, α	303,5776346 gon
Zenith angle, z_v	102,459666 gon
Horizontal distance, S_H	9270,1001 m
Ell height difference, ΔH = z	-358,3410 m

Variance-covariance propagation law: $\mathbf{C}_{(\alpha,s,z_v)} = \mathbf{F} \cdot \mathbf{C}_{LG} \cdot \mathbf{F}^T$ (11.14)

Here we use ΔH and not the zenith angle, z_v.
Mathematical connections give the transformation matrix **F**, azimuth (α), horizontal distance (S_H) and ΔH are used as observations:

$$\begin{bmatrix} d\alpha_{12} \\ dS_{H12} \\ d\Delta H_{12} \end{bmatrix} = \mathbf{F} \cdot \begin{bmatrix} dx_{12} \\ dy_{12} \\ dz_{12} \end{bmatrix} = \begin{bmatrix} -\sin\alpha_{12} \cdot \rho / S_{H12} & \cos\alpha_{12} \cdot \rho / S_{H12} & 0 \\ \cos\alpha_{12} & \sin\alpha_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} dx_{12} \\ dy_{12} \\ dz_{12} \end{bmatrix} \quad (11.13a)$$

Calculating the transformation matrix **F** by using equation (11.13), denomination is mm and mgon. Equation (11.13) gives the mathematical connection between the observables in the local geodetic system (LG) and the corresponding derived observations in the map projection plane (horizontal distance, azimuth, height difference or zenith angle. Here we use ΔH and not the zenith angle, z_v.

The transformation matrix **F** for use in the error propagation law (variance/co-variance propagation law):

$$\mathbf{F}_{\alpha S,\Delta H} = \begin{pmatrix} 0,006856612 & 0,000385730 & 0 \\ 0,056167778 & 0,998421344 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Calculating the variance/co-variance matrix **C** for the derived observations bearing, distance and ΔH by using equation (11.14):

$$\mathbf{C}_{\alpha,S,\Delta H} = \mathbf{F} \cdot \mathbf{C}_{\mathbf{LG}} \cdot \mathbf{F}^{\mathbf{T}} = \begin{pmatrix} 0,000009 & 0,000105 & 0,000924 \\ 0,000105 & 0,153272 & 0,018014 \\ 0,000924 & 0,018014 & 1,069399 \end{pmatrix}$$

Calculating the standard deviations on azimuth (α), horizontal distance (S_H) and ΔH of the baseline from the variances in the main diagonal, still very low values of the standard deviations:

S_α = 0,003 mgon S_S = 0,39 mm $S_{\Delta H \text{ Ellipsoidal}}$ = 1,03 mm

Comment: The standard deviations on the bearing ($\varphi = t$), distance in the UTM map projection plane (D_{MP}) and the NN2000 ΔH are the same:
$S\varphi$= 0,003 mgon S_D = 0,39 mm $S_{\Delta H \text{ NN2000}}$ = 1,03 mm
Reason: The standard deviations of corrections are zero, and therefore the error propagation law gives the same standard deviations on for instance the azimuth and the bearing.

(d) Often empirical values are used for the standard deviations.
Use assumed empirical values of the GPS observations and calculate the variances of the distance, bearing and the zenith angle of both of the baselines.

State with equations how the complete variance/covariance matrix can be calculated.

Evaluate the methods and the results of the 2 calculations in (a) and (b).
How will you calculate the optimal accuracy measures of the measured baselines?

Not covered in details this year, as the method has some serious weaknesses, and therefore is normally not used in the adjustments. See chapter 11.4.2 for more

The starting point is the empirical accuracy of the base lines, for instance:
Horizontal distance: 5 mm + 1 mm/km
Height difference: 7,5 mm + 1,5 mm/km

From these standard deviations we can calculate the variance / standard deviation of the bearing, distance and the height difference.
We obtain more realistic values, but the problem is to give good values of the co-variances. This empirical method does not consider possible variations of geometry, noise, equipment for the various baselines. This is accounted for in the method in question 2.

The too low values in question 2 can be adjusted by scaling up the accuracy measures like the standard deviations with a factor of 10 (is regularly done for some of the GNSS equipment). Thus the elements in the variance-covariance matrix must be multiplied with $10^2 = 100$.

Task no. 3. Checking the results in 1 and 2 with the program GISLINE

(a) The calculations above can be tested by using the baseline transformation program in for instance the Norwegian land surveying programs GIS/Line or Gemini GPS/NET.

Is done, gave the same results…
See also the GISLINE manual in the compulsory assignment no. 1 covering this item.

The documentation file (the report) from Gisline is added to Blackboad. But, no comments, and it is in Norwegian. Some results also found in this solution, is highlighted.